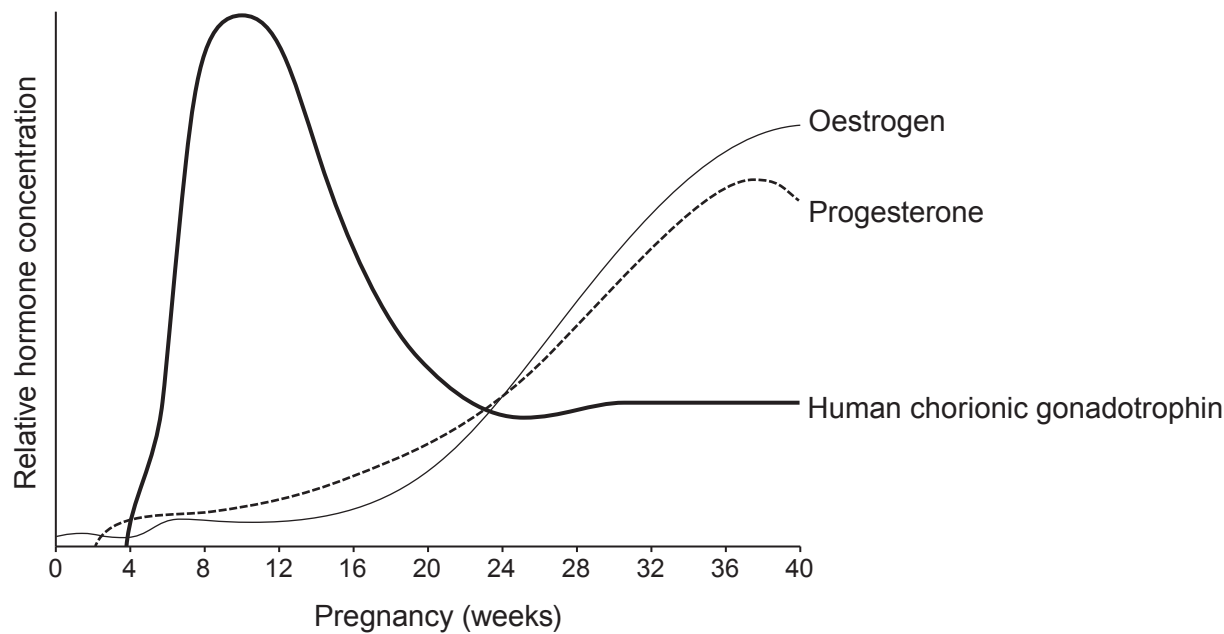


2. Human chorionic gonadotrophin (hCG) is secreted by the developing embryo and is crucial in maintaining early pregnancy. **Graph 2.1** shows how levels of hCG change over time, together with two other hormones which have important roles in pregnancy.

Graph 2.1



- (a) With reference to **Graph 2.1**, explain why hCG levels are used in pregnancy tests to determine if a woman is pregnant. [2]

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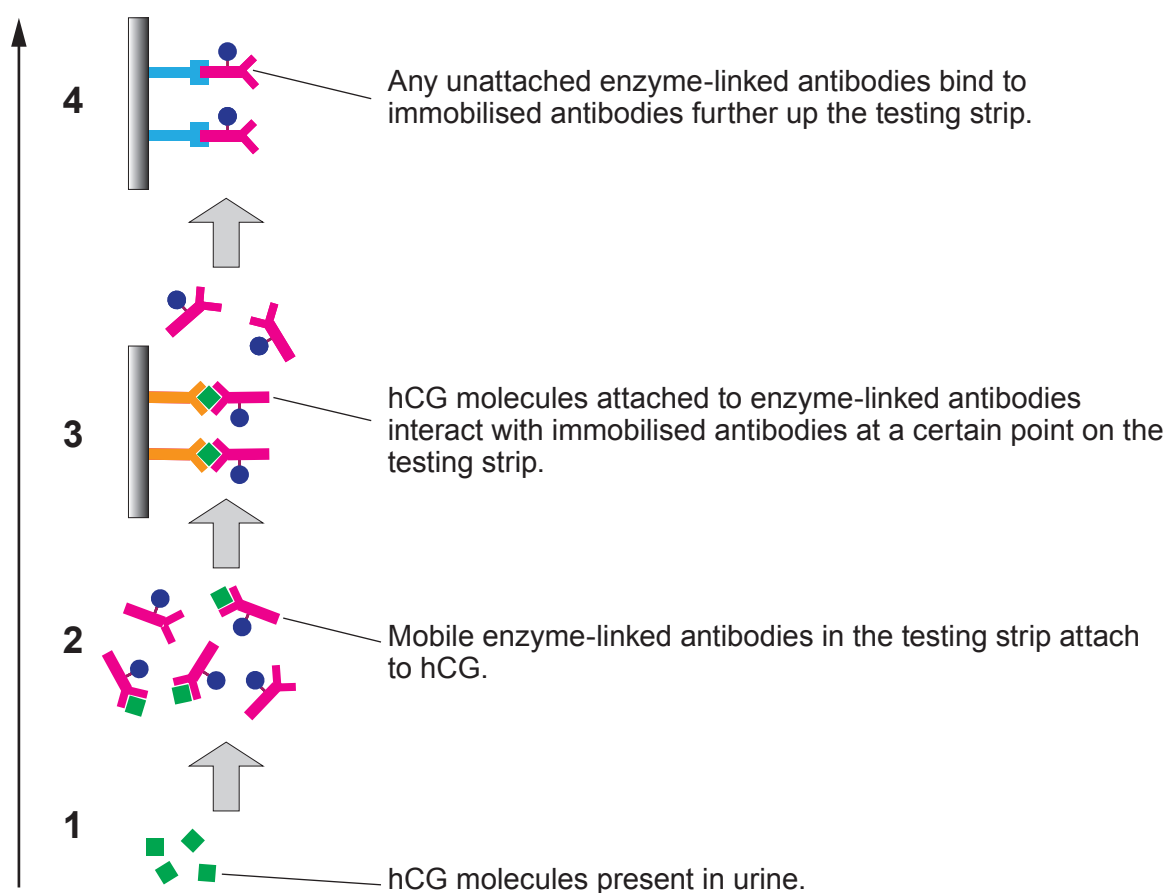
- (b) One type of pregnancy test uses mobile proteins, called antibodies, which attach to hCG. These in turn are attached to an enzyme that changes the colour of a dye in an absorbent testing strip.

Another set of antibodies specific to hCG are immobilised to the colourless dye in the strip. If hCG is present in urine, it will interact with both sets of antibodies (forming an antibody sandwich). When both sets of antibody are bound to hCG, the enzyme can bind with the dye, changing its colour.

Immobilised antibodies further up the testing strip will bind any unattached enzyme-linked antibodies, changing their colour, functioning as a control.

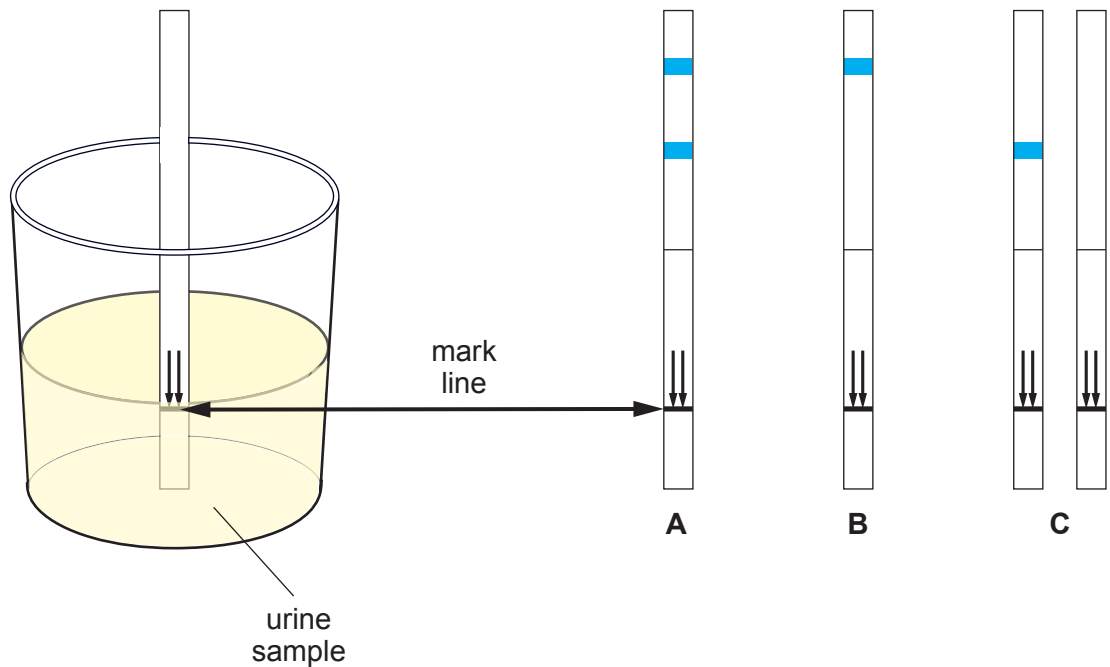
This is summarised in **Image 2.2**.

Image 2.2



The set of testing strips in **Image 2.3** show a number of different possibilities.

Image 2.3



(i) Determine which of the tests **A** or **B** is positive for a pregnancy and explain your reasoning. [3]

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(ii) Explain why both of tests **C** are invalid. [2]

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(c) State the function of hCG in pregnancy and what happens if levels of hCG fall or are too low at the start of the pregnancy. [3]

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(d) hCG is a protein coded for by two genes. Sometimes mutations occur which may affect the structure and functionality of the hormone. Explain why mutations in the hCG genes may or may not effect pregnancy. [2]

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(e) With reference to **Graph 2.1**, explain the functions of the hormones oestrogen **and** progesterone during pregnancy. [4]

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